

Coordinated maintenance. Explainable block plans.

Illustrated technical companion to the six-slide SIH submission.

The decision we support

Choose which maintenance work can fit into available railway blocks, coordinate compatible departmental activities, and explain what cannot be scheduled. Retain human review and external operational authority.

What is inside

22 diagram pages cover context, architecture, data, security, monthly and weekly planning, compatibility, verification, recovery, deployment and feedback. The closing pages explain evidence and judge preparation.

What is established

The design research, supplied problem statement and architecture are documented. The team reports an existing prototype; its functionality has not been independently verified for this submission.

What remains to prove

Authorized system access, model correctness, data quality, operational usability, solve-time scaling and measured improvement need testing. The worked example is illustrative.

How to use this document

Use the six-slide PDF for the portal after filling the Team ID and demo link. Use this companion for technical discussion, together with the full 98-page production blueprint and the research/judge guide.

The complete planning argument

Connect each requirement to a mechanism and a demonstration.

1 / Integrate the inputs

TMS, SMMS and TDMS maintenance records plus timetable and Control Office corridor information enter a shared, versioned contract. Begin with approved file imports; production interfaces require authorization.

2 / Prioritize and schedule

Use criticality, due dates and asset impact as transparent priorities. CP-SAT places work within hard constraints. A future duration predictor needs authorized historical outcomes and validation.

3 / Coordinate complete packages

Include access, setup, task dependencies, resources, testing and restoration. Allow parallel tasks only when compatibility is established. Independently check the final assignment.

4 / Plan across horizons

A monthly allocation provides the parent programme. A weekly proposal reconciles commitments with current readiness and capacity. Changed conditions trigger a traceable amendment or replan.

5 / Show proof, not just screens

Demonstrate a feasible package, an incompatible pair, an overlong package, an urgent new task and a stale input revision. Show the checker result, why-not explanation and plan diff.

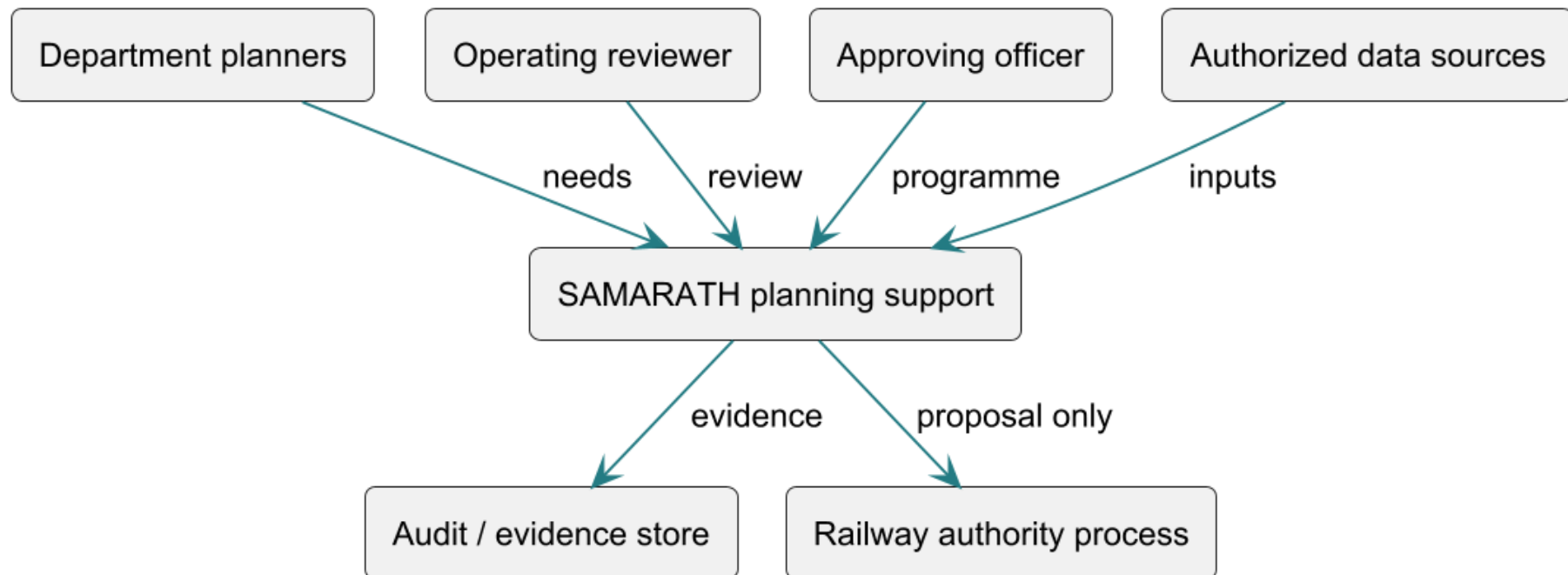
Key terms

TMS: Track Management System. SMMS: Signalling Maintenance & Management System. TDMS: Traction Distribution Management System. COA: Control Office Application. CP-SAT: constraint programming solver. RLS: row-level security.

D01 / System context

SAMARATH consumes approved information and proposes programmes. It does not grant or release operational blocks.

D01 - System context



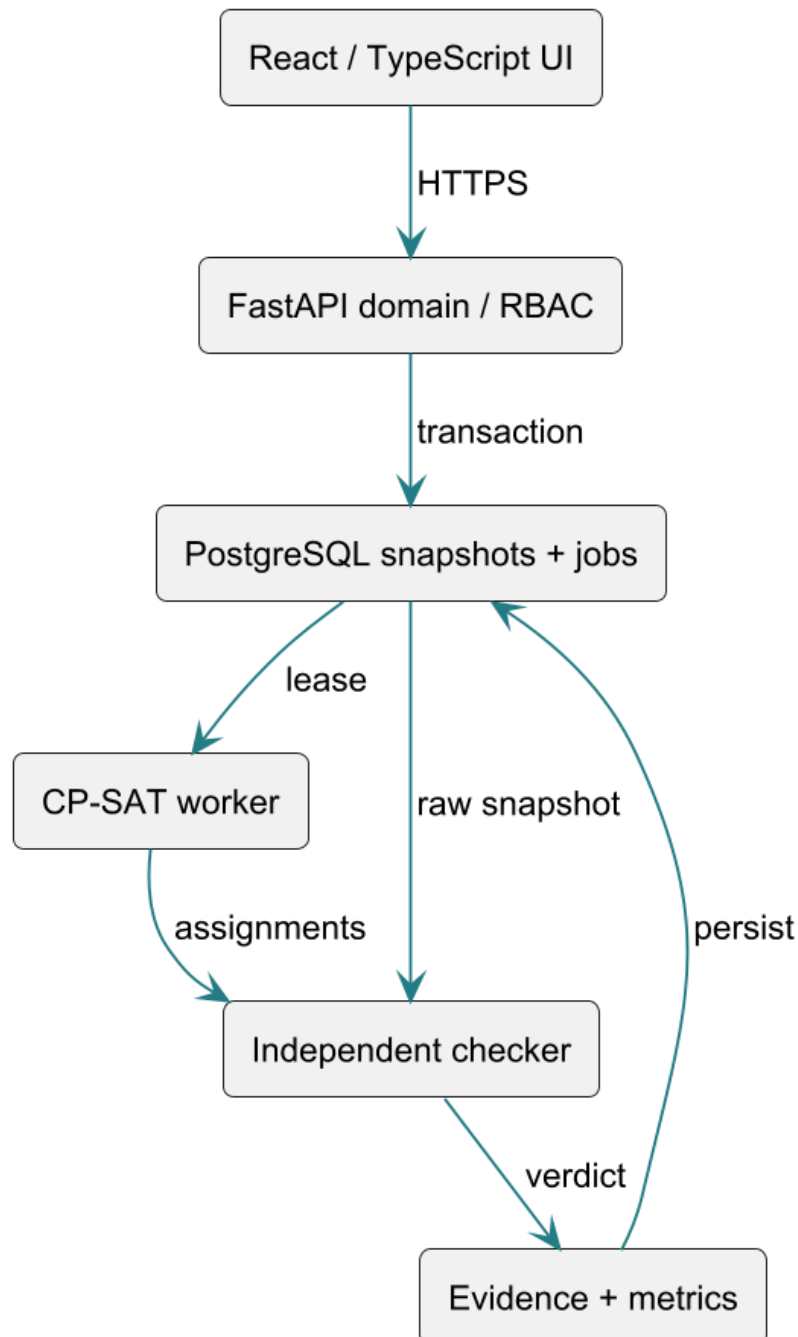
A proposal is not permission to occupy the railway.

Keep departmental preparation, operating review and programme approval distinct. Operational grant, protection and release remain with the authorized railway process.

D02 / Core components

The checker independently interprets raw inputs. It does not import optimizer predicates.

D02 - Core components



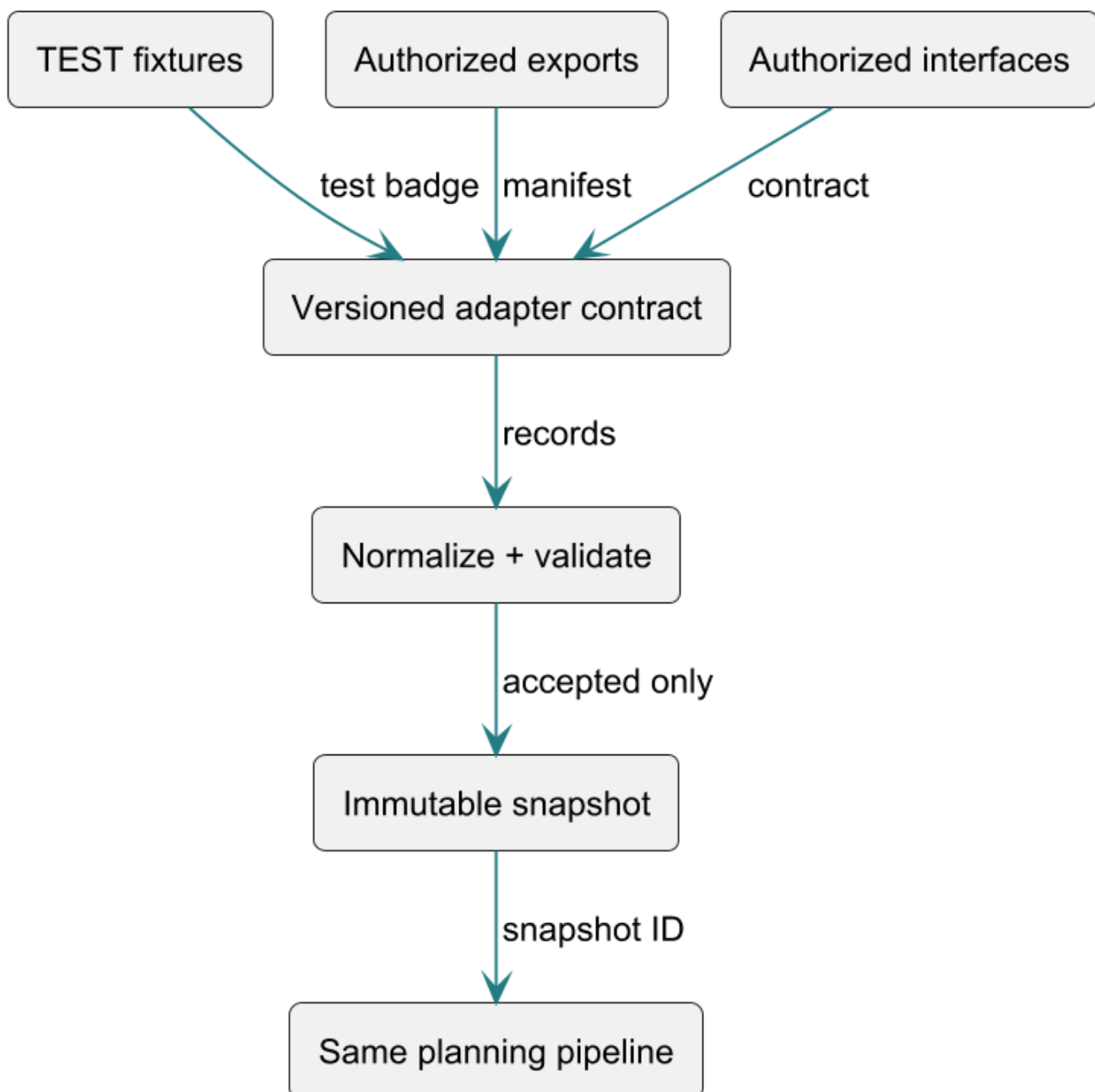
The checker must challenge the solver output.

Recompute feasibility from the raw snapshot and the published assignment. Sharing the same derived candidate data can reproduce the same error in both components.

D03 / Three input modes

Adapter choice changes input provenance and access controls. Planning logic is shared.

D03 - Three input modes



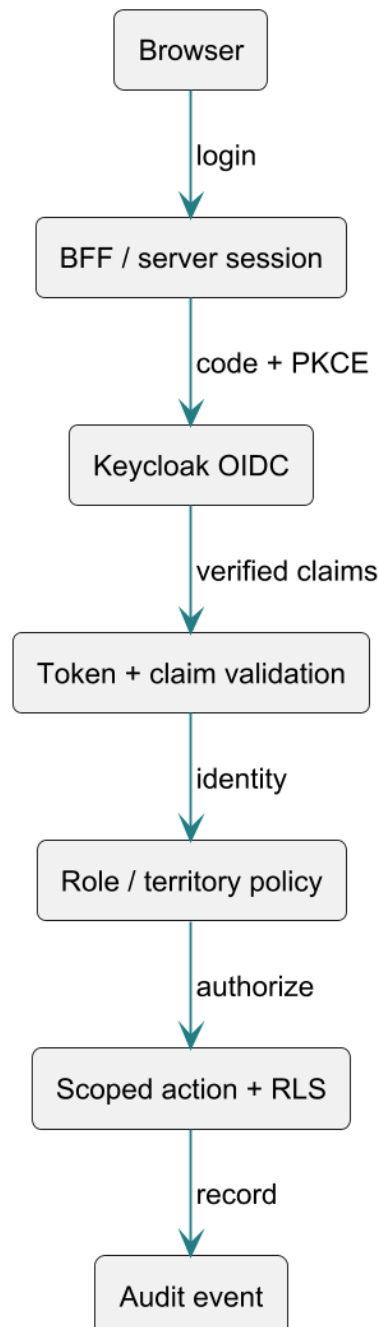
One contract, three evidence levels.

Label synthetic TEST data visibly. Import approved exports before attempting live integration. A working CSV adapter does not establish access to CRIS production systems.

D04 / Login and scope

Server-derived roles and territory scope protect every request; browser-supplied role names are not authority.

D04 - Login and scope



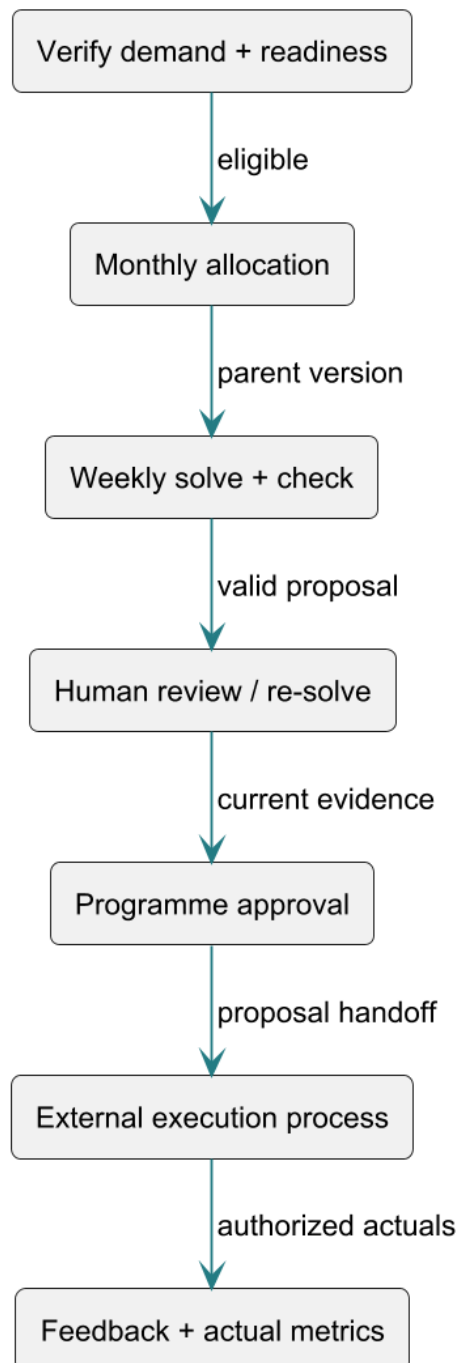
Every action needs both a role and a territory.

Authentication alone is insufficient. Enforce scope in the API and database, audit changes, and test that users cannot read or approve plans outside their authority.

D05 / End-to-end planning workflow

Programme approval is a SAMARATH review state; operational authority remains external.

D05 - End-to-end planning workflow



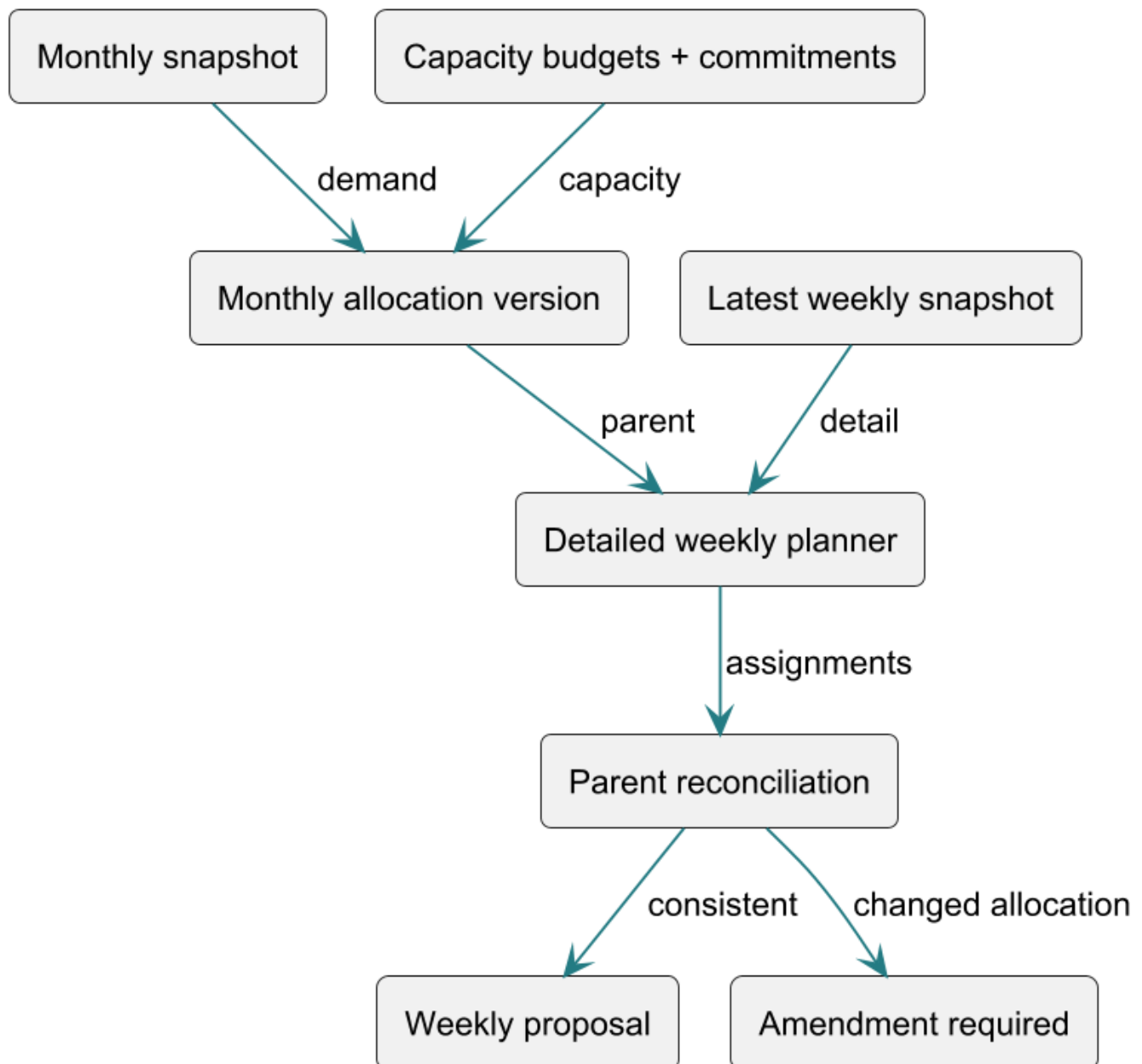
Readiness is part of feasibility.

Check crews, materials, equipment, permissions, testing and restoration before offering a package. Required work that cannot fit must remain visible for escalation.

D06 / Monthly-to-weekly consistency

Monthly capacity feasibility is provisional. Changes outside the parent allocation require an explicit amendment.

D06 - Monthly-to-weekly consistency

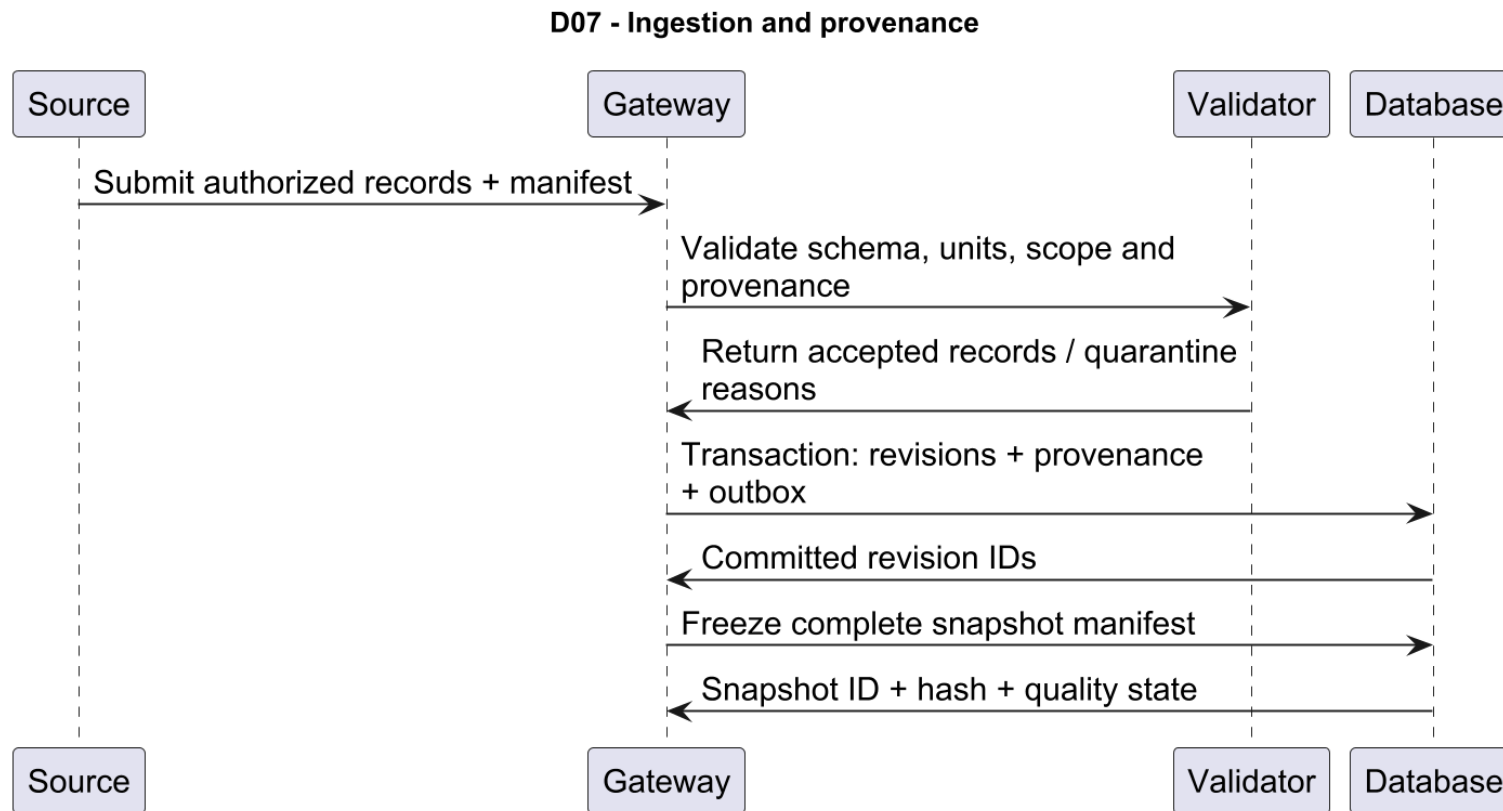


The weekly programme must explain changes to the monthly plan.

Use a named monthly parent version. Reconcile completed, deferred, newly urgent and outstanding work; do not silently replace the monthly commitments.

D07 / Ingestion and provenance

Invalid records remain quarantined and traceable; they cannot silently enter an eligible snapshot.

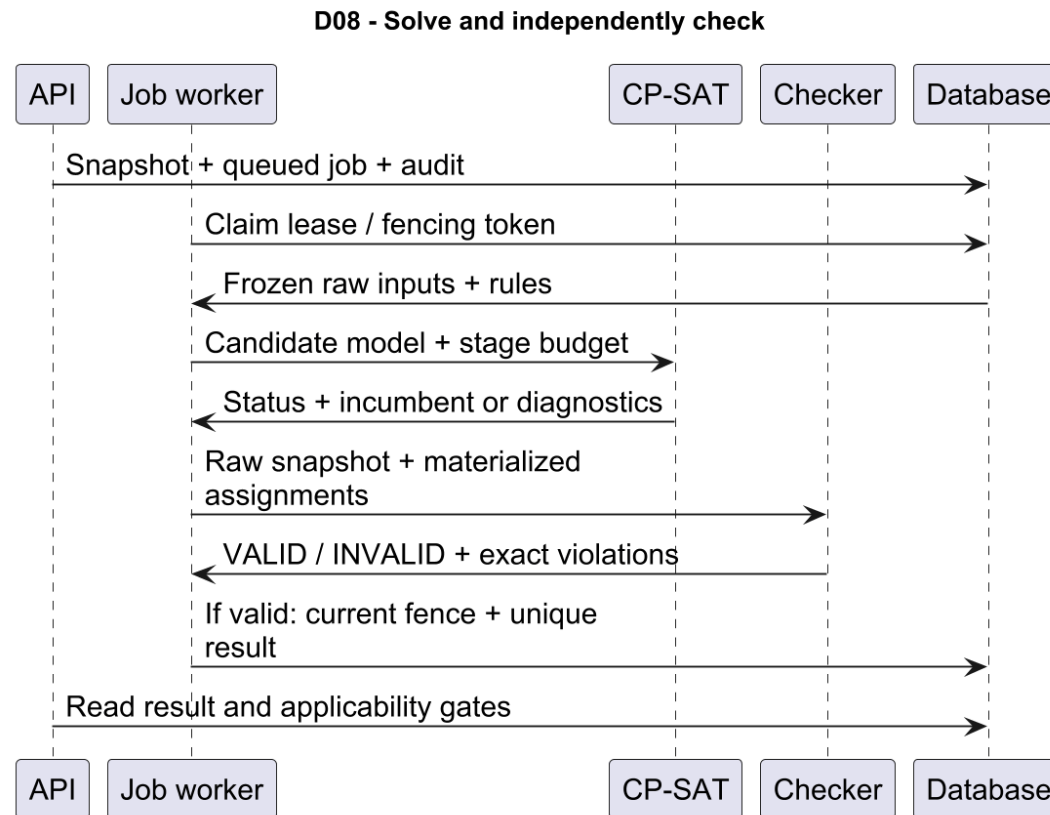


Bad inputs must not silently become valid planning facts.

Validate units, identifiers, geography, time zones and source freshness. Quarantine ambiguous records. Preserve the source revision and snapshot hash for replay.

D08 / Solve and independently check

A solver incumbent is not publishable until the checker passes and currentness/approval gates also hold.

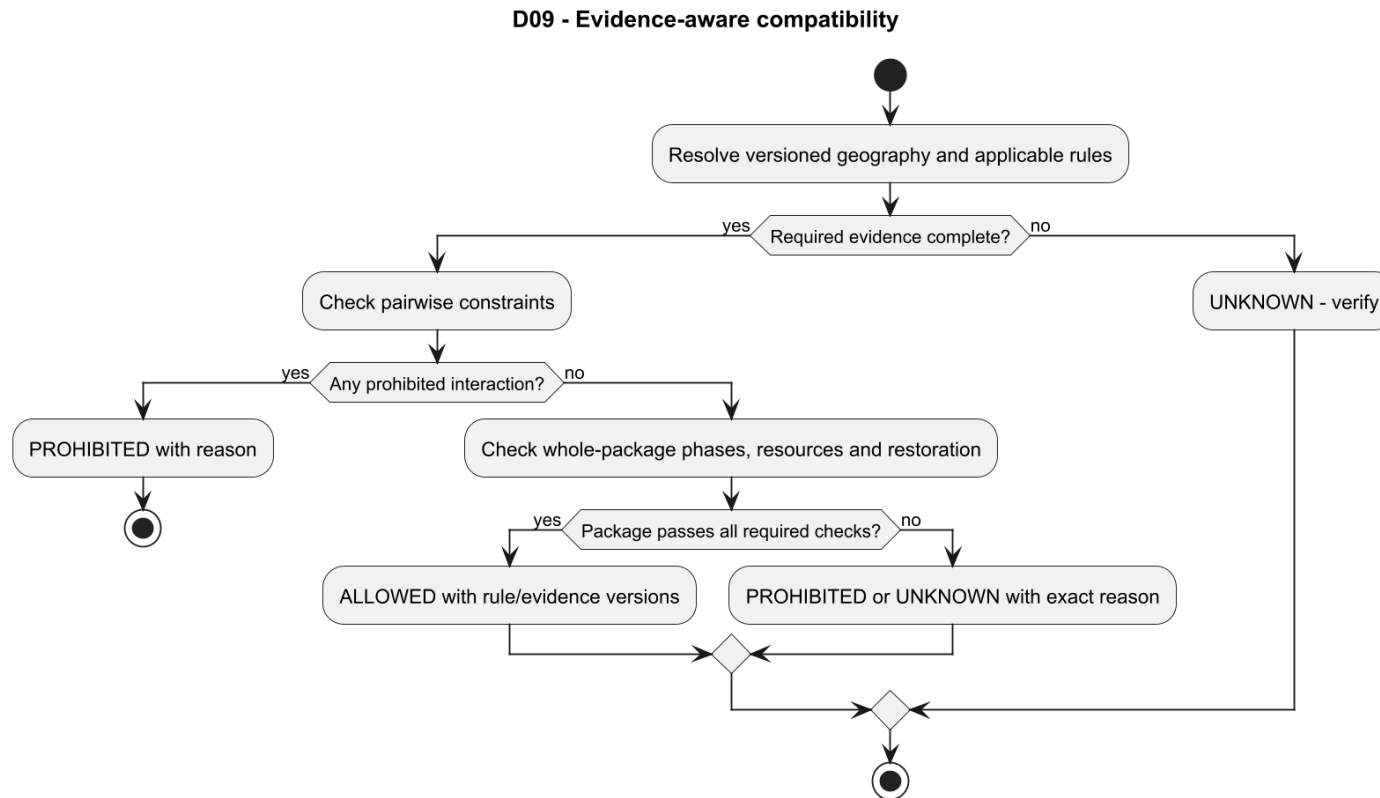


Report what the solver actually established.

OPTIMAL, FEASIBLE, INFEASIBLE and timeout/UNKNOWN are different outcomes. No incumbent means no executable proposal. Only a current, independently checked result can be published.

D09 / Evidence-aware compatibility

Missing evidence yields UNKNOWN. Pairwise permission is necessary but not sufficient for a complete package.



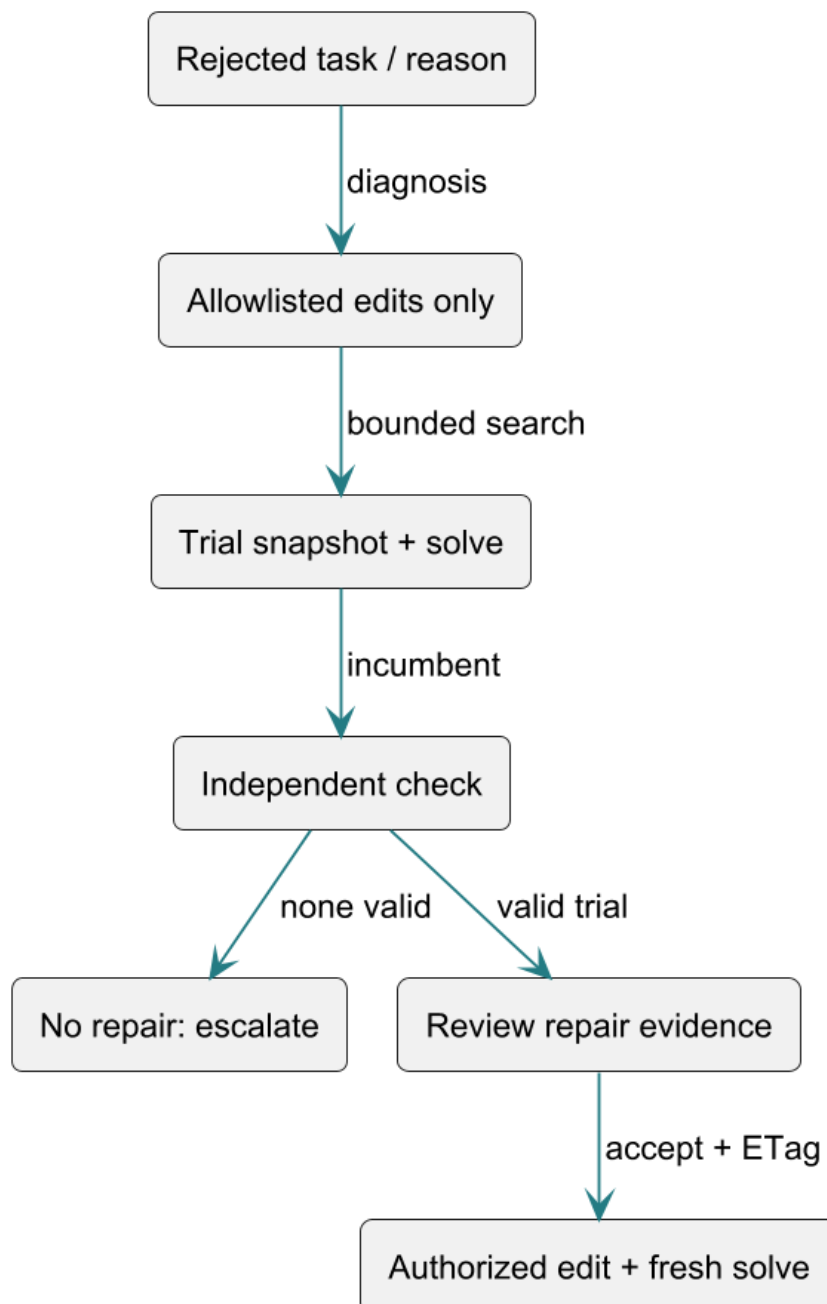
Unknown compatibility cannot authorize simultaneous work.

A shared location is not enough to bundle tasks. Check isolation, access, phase overlap, protection and restoration; retain the evidence and rule version behind each decision.

D10 / Permitted repair

A repair changes inputs or proposes an authority request; it never relaxes protection or grants itself access.

D10 - Permitted repair

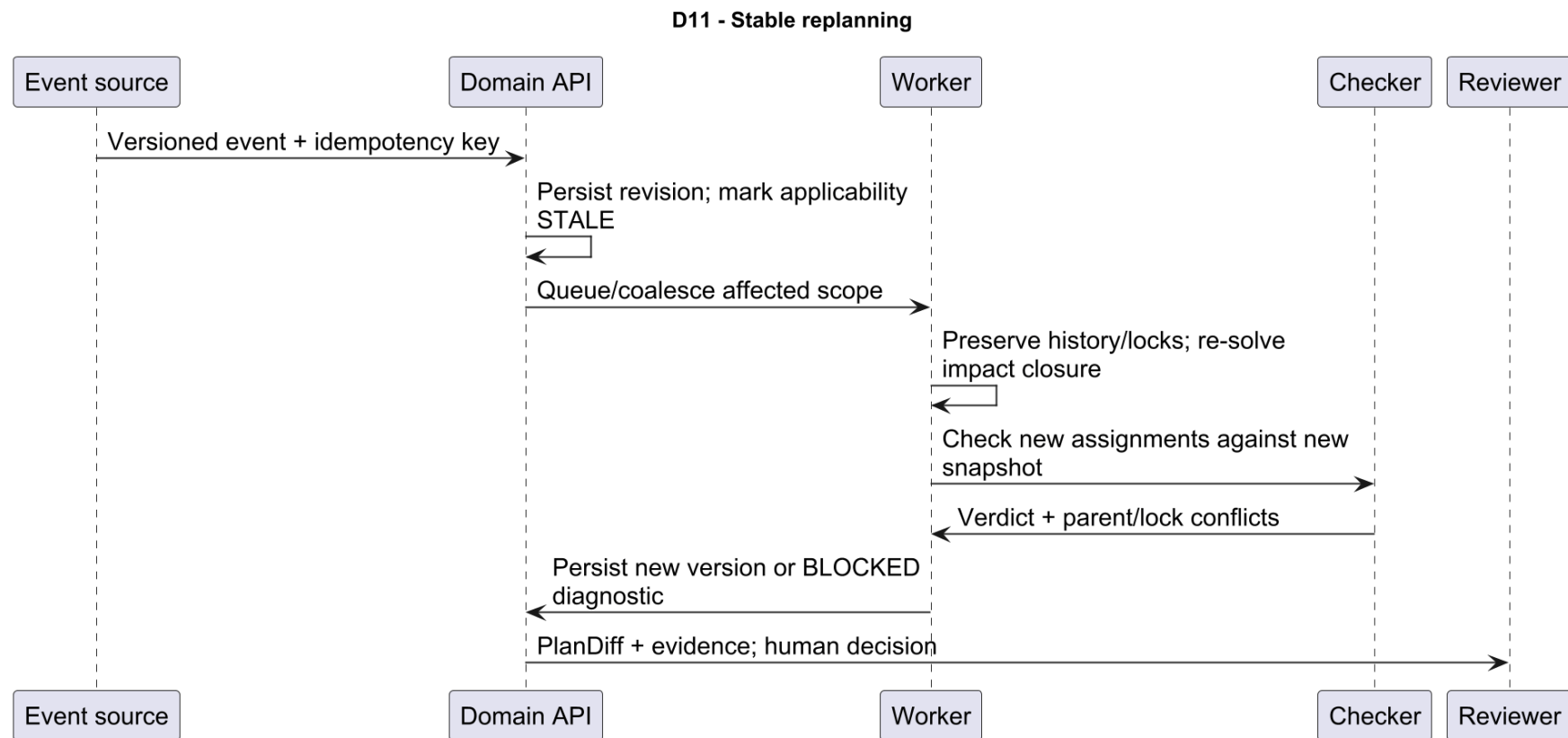


An explanation should lead to a permissible next action.

Test only authorized changes, such as another available window or resource. Never suggest relaxing a safety requirement. A trial repair remains a proposal until accepted and rechecked.

D11 / Stable replanning

History and operational commitments persist. A conflicting new event blocks applicability; it does not unlock work.



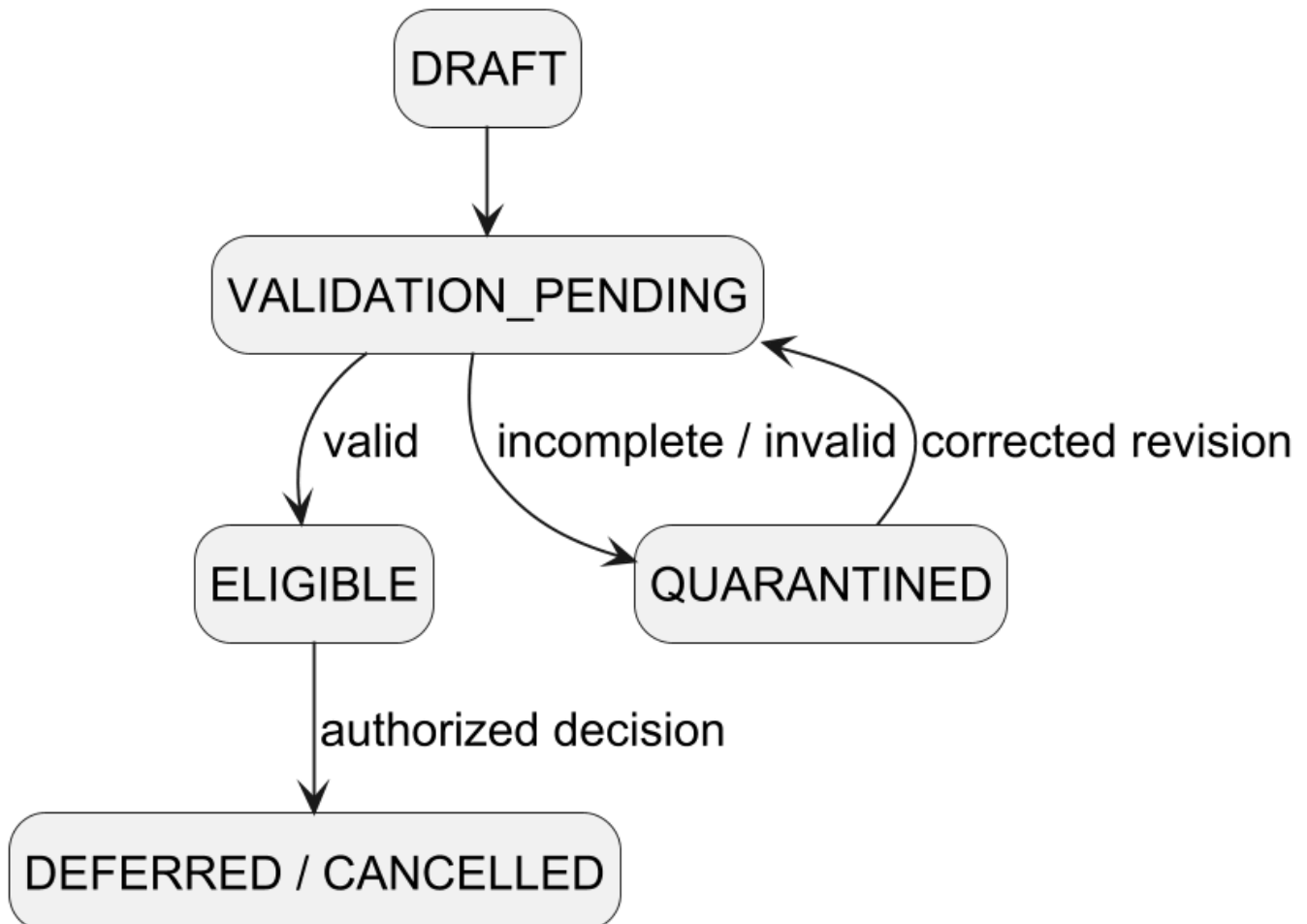
Recover with the fewest justified changes.

Protect required coverage and hard locks, then minimize changed assignments before improving efficiency. Invalidate stale results when inputs change and expose a before/after plan diff.

D12_task / Task lifecycle

Eligibility is earned through validation.

D12 - Task data



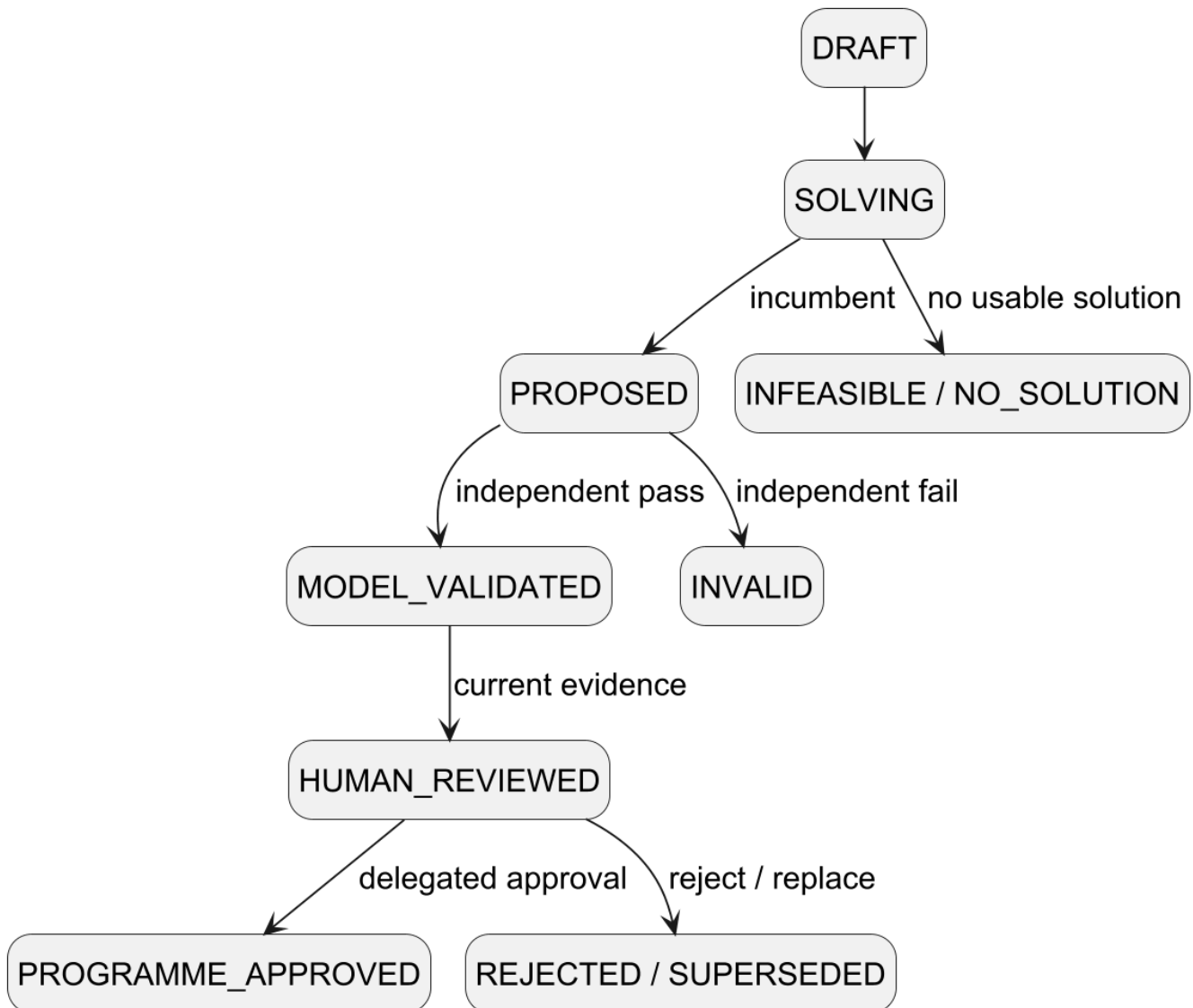
Eligibility is earned through validation.

A correction creates a new revision. Deferral and cancellation require authority and a reason. Keep residual work visible instead of losing it through a status change.

D12_plan / Plan lifecycle

Validation and human approval are separate gates.

D12 - Plan version



Validation and human approval are separate gates.

The diagram compresses the unsuccessful-solve branch. The runtime must distinguish proven infeasibility from timeout or unknown status. Applicability to current inputs is a separate property.

D12_authority / External operational authority

Execution states are authorized observations.

D12 - Railway authority mirror



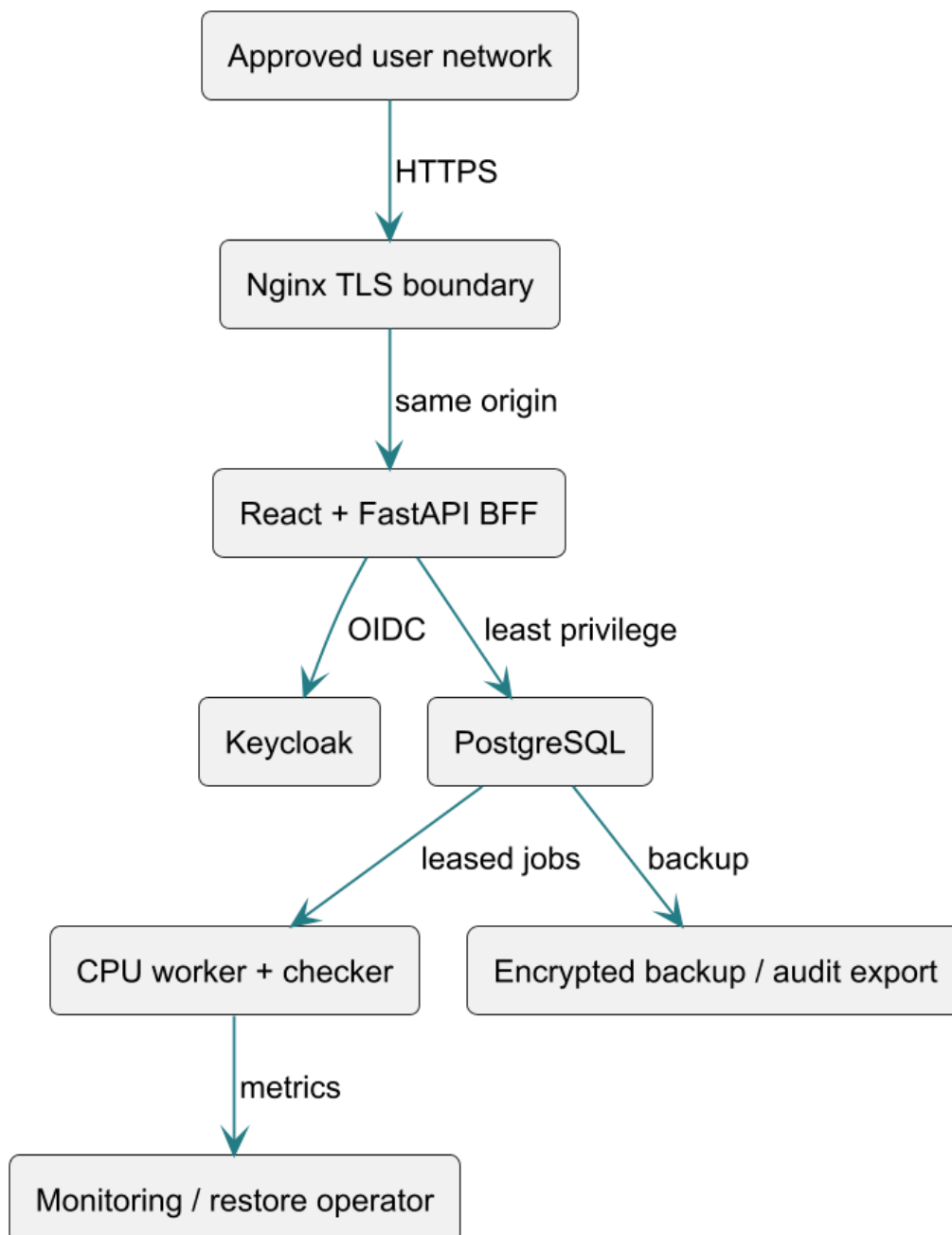
Execution states are authorized observations.

SAMARATH records reported operational states. It does not grant a block or release an isolation. Unknown or missing observations must remain explicit.

D13 / On-premises deployment

Development/demo uses Compose. Pilot redundancy and disaster recovery are acceptance work, not assumed properties.

D13 - On-premises deployment



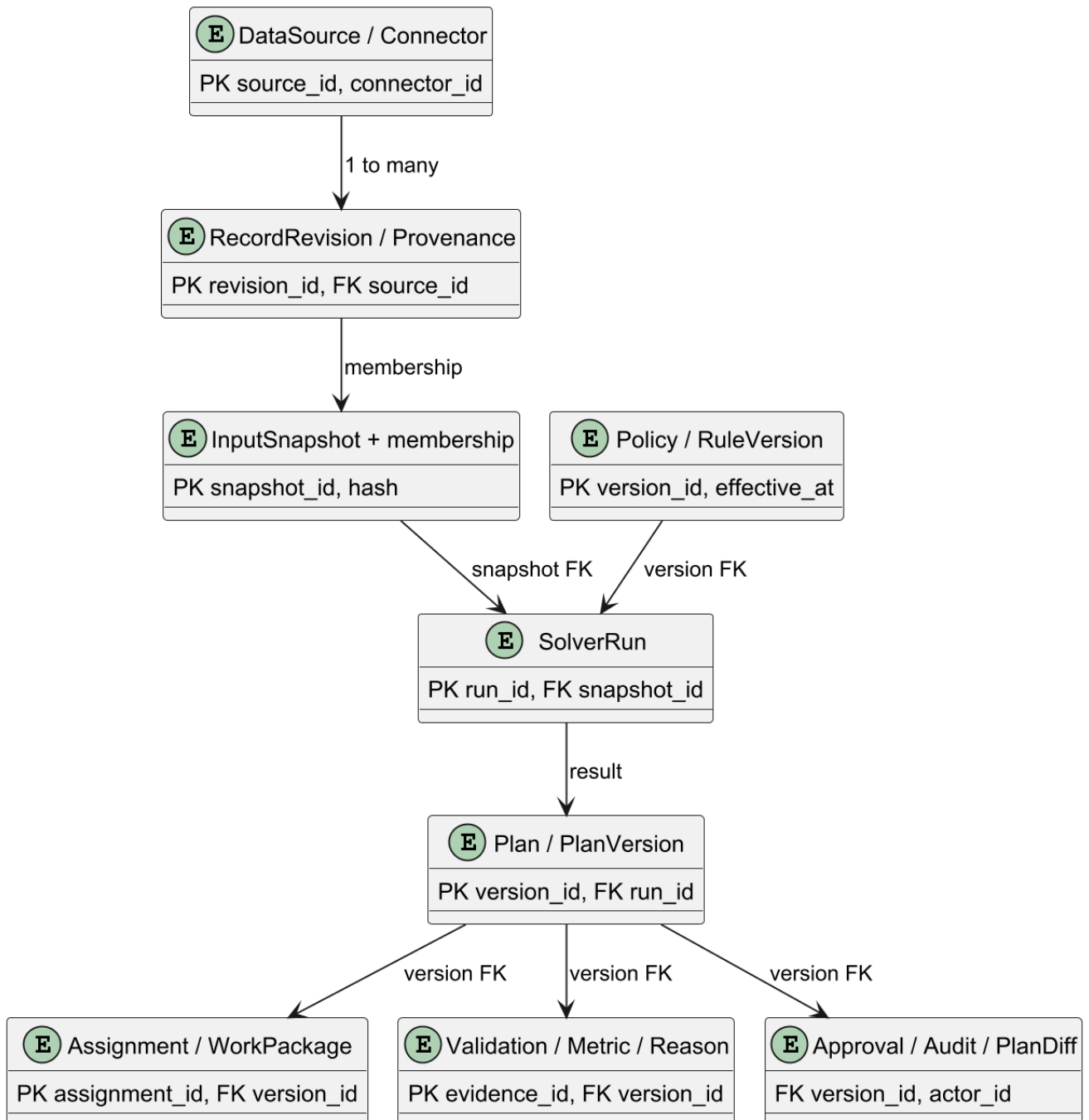
Start with a bounded, supportable deployment.

Separate CPU-bound solving from web requests. Monitor worker health, backups, storage and access. Deployment topology is a proposed pilot design, not a production approval.

D14 / Relational model: planning spine

This view shows the main foreign-key spine. The full field catalogue and supporting resources/rules are in Section 13.

D14 - Relational planning spine



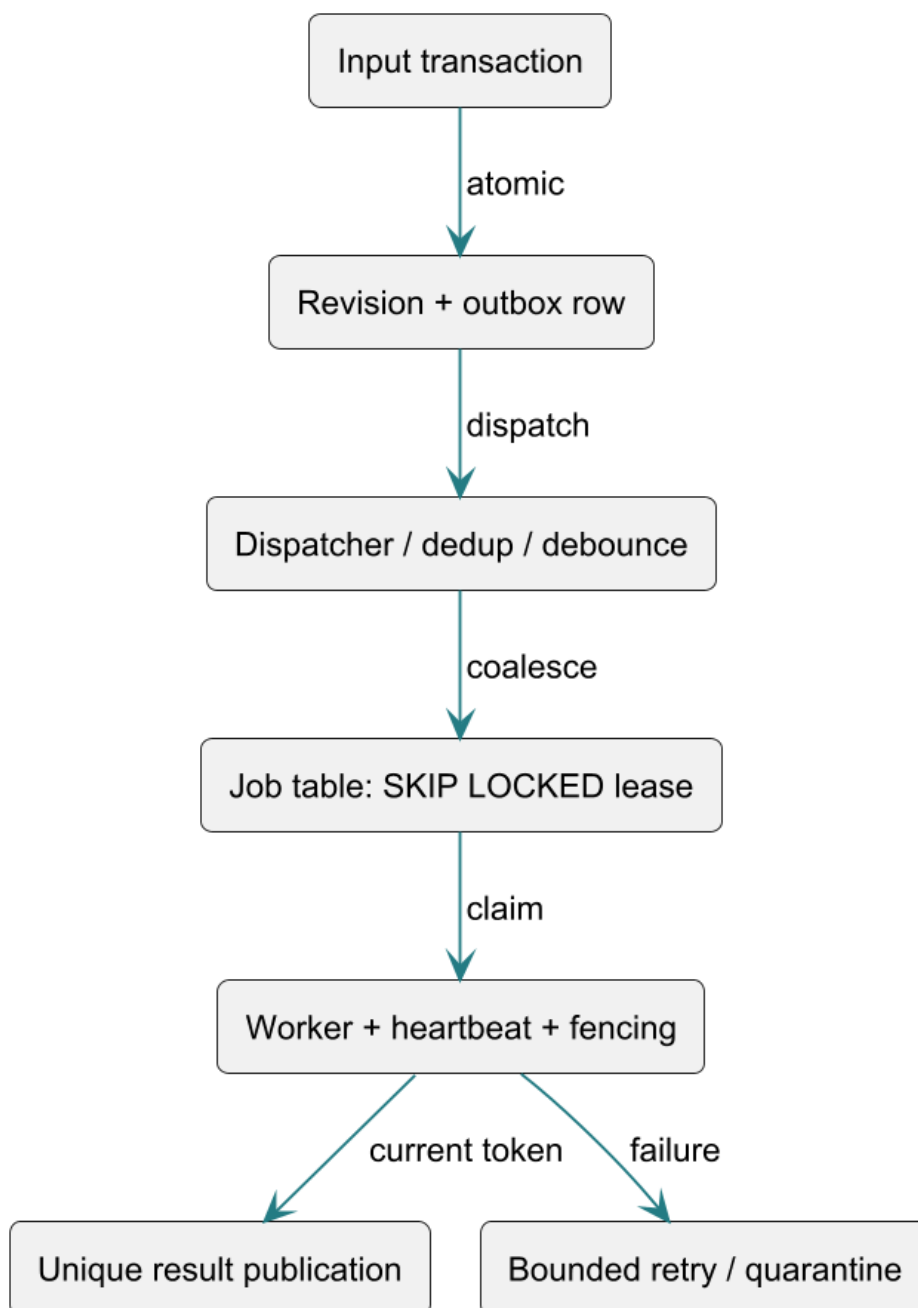
Every displayed plan should have a reproducible evidence chain.

Link the input snapshot, rule and policy versions, solver outcome, checker report and approval record. Never overwrite the history needed to explain a past decision.

D15 / Durable event and job processing

At-least-once delivery requires idempotency. A lease and fencing token prevent an obsolete worker from publishing.

D15 - Durable event and job processing



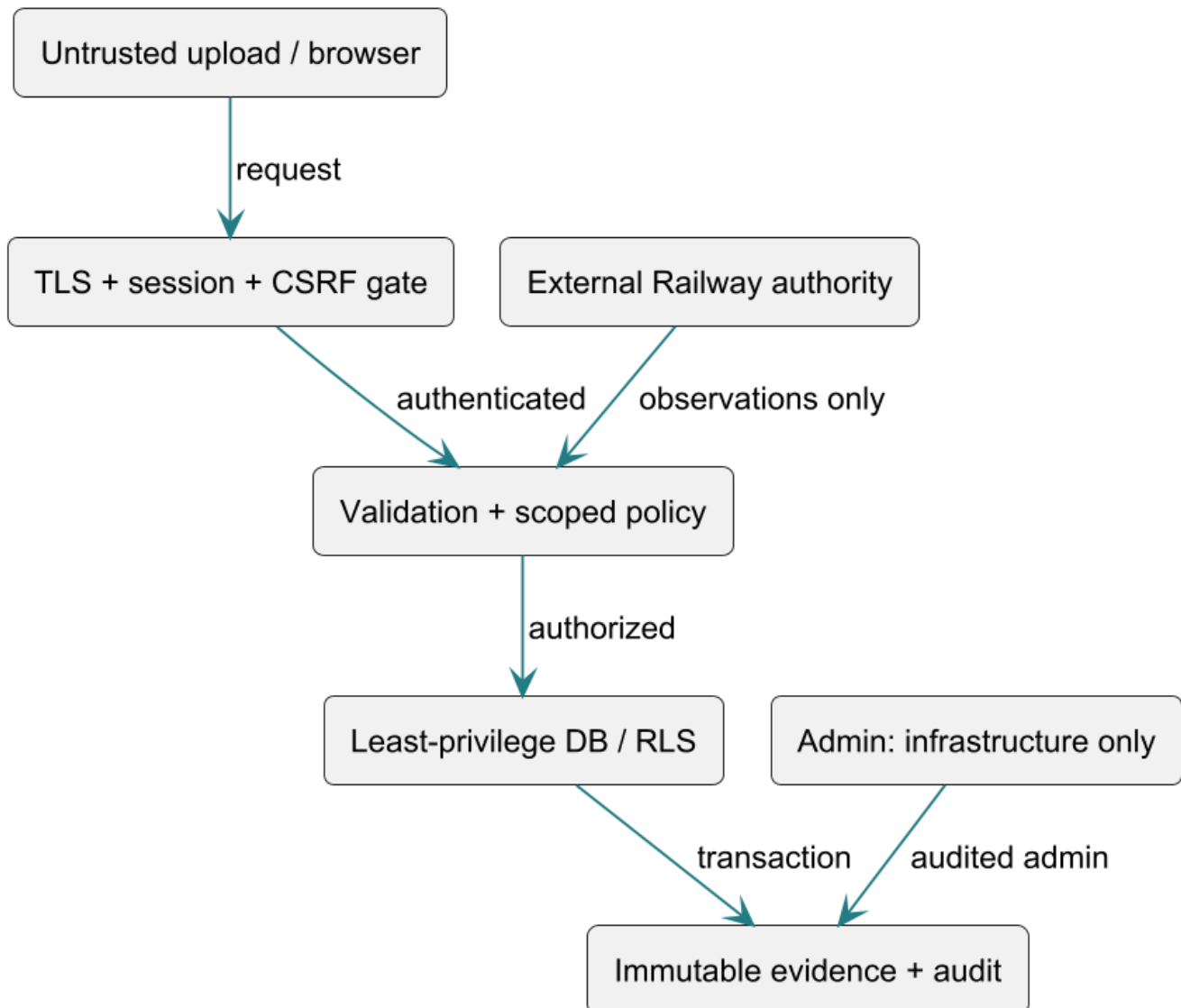
A retry must not publish a duplicate or stale plan.

Use durable transactions, leases, heartbeats and fencing tokens. Idempotent publication and controlled retries matter when workers crash or jobs overlap.

D16 / Trust and security boundaries

Domain authority, software administration and operational railway authority remain separate.

D16 - Trust and security boundaries



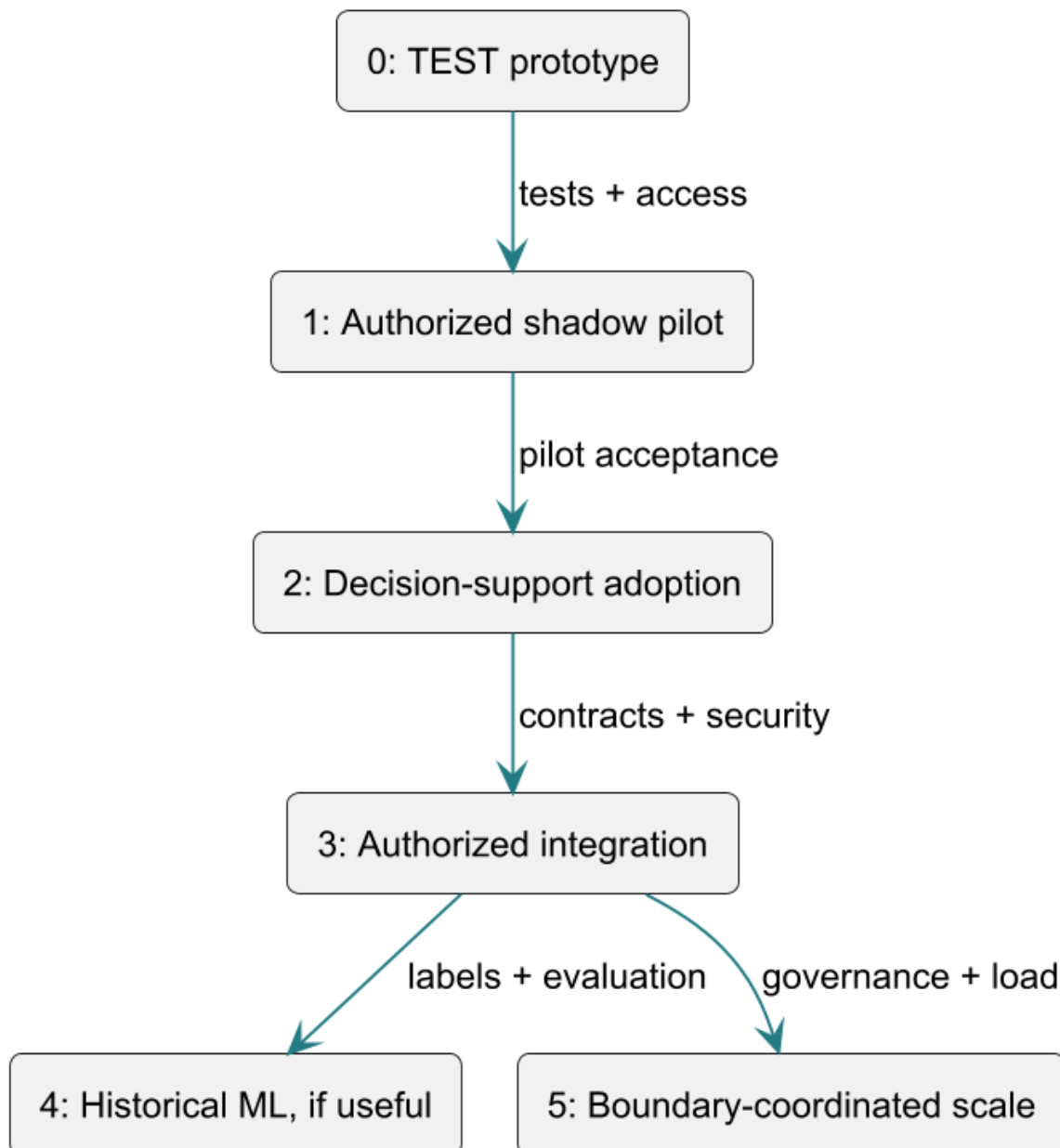
Separate infrastructure access from railway decision authority.

Protect sessions and uploads, use least privilege, enforce territory scope, and retain audit evidence. A system administrator should not automatically inherit programme approval rights.

D17 / Capability evolution

Progress is gated by evidence. ML and multi-division coordination are separate optional workstreams after the core is accepted.

D17 - Capability evolution



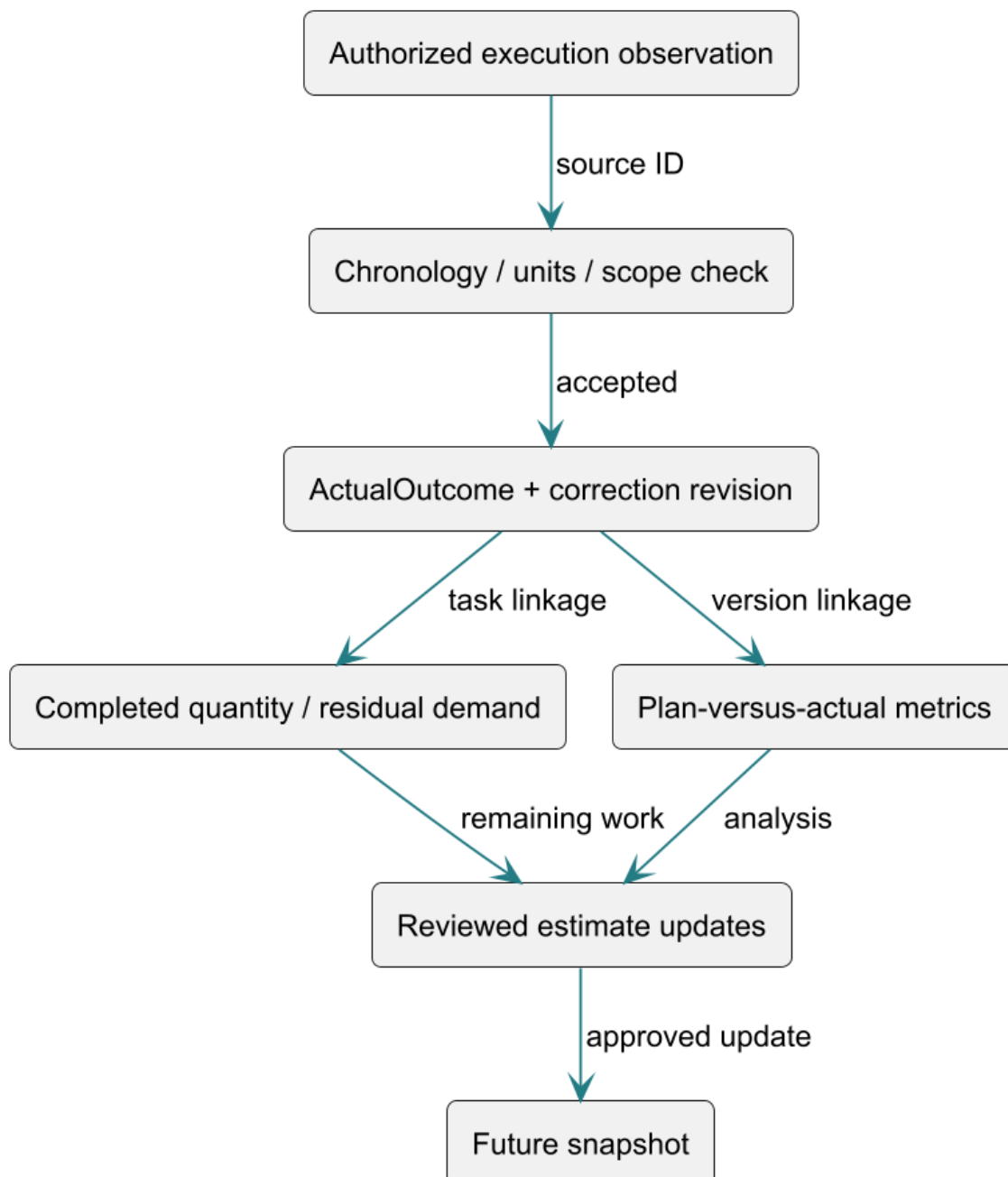
Scale through measured gates.

Begin with a fictional corridor, then an authorized shadow pilot. Historical ML is data-gated. Larger territories require boundary coordination, sparse candidates and benchmark evidence; national scale is not yet demonstrated.

D18 / Execution feedback loop

Actuals inform estimates only with traceable corrections. A release does not automatically prove all planned work is complete.

D18 - Execution feedback loop



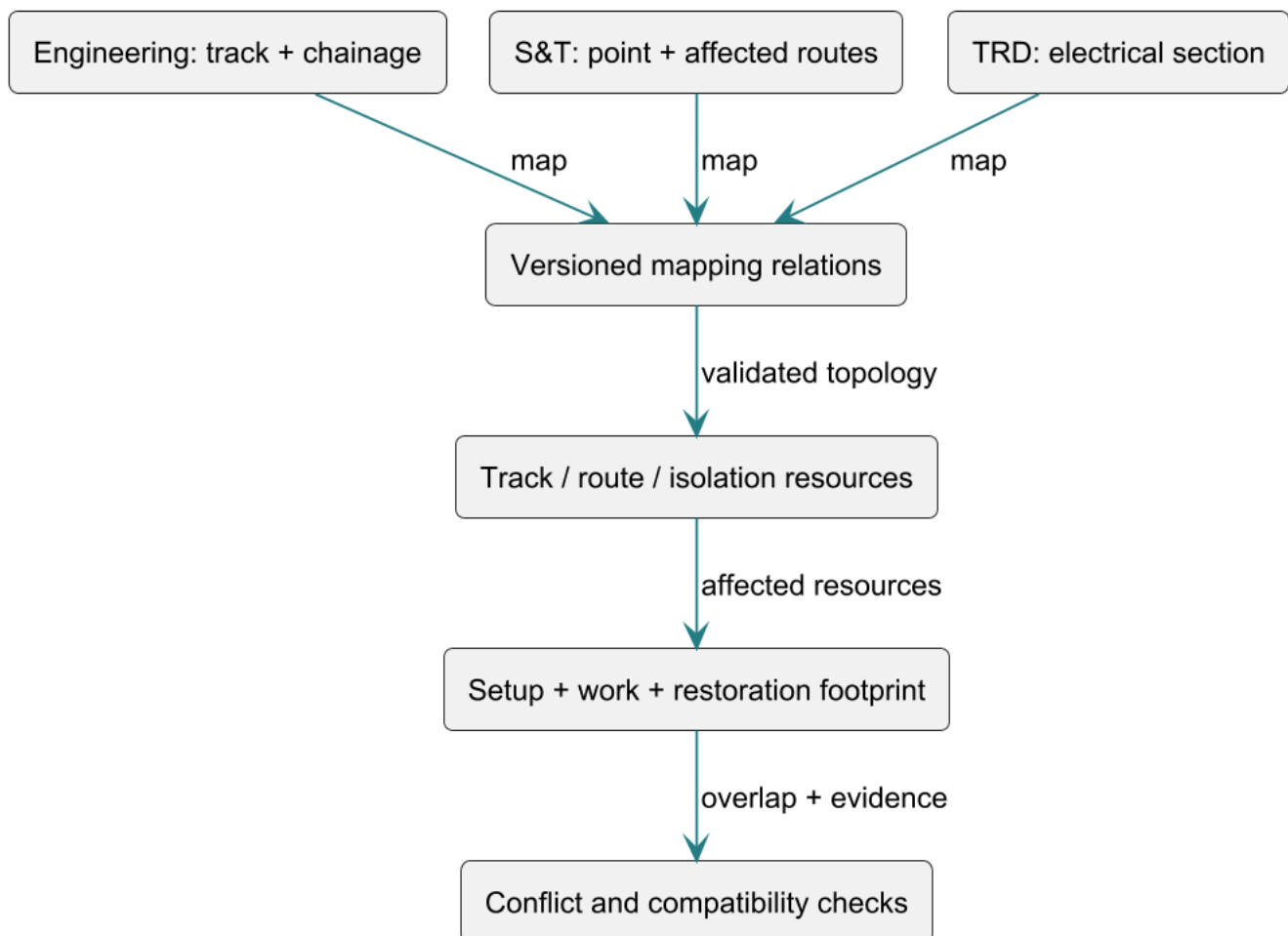
Learning starts with trustworthy execution feedback.

Validate actual chronology and quantities. Reconcile partial completion and residual demand. Review estimate changes before they affect a new snapshot; do not silently train on unreliable outcomes.

D19 / Cross-domain geography

Map all domains onto versioned occupation resources. Coincident station names alone do not establish compatibility.

D19 - Cross-domain geography



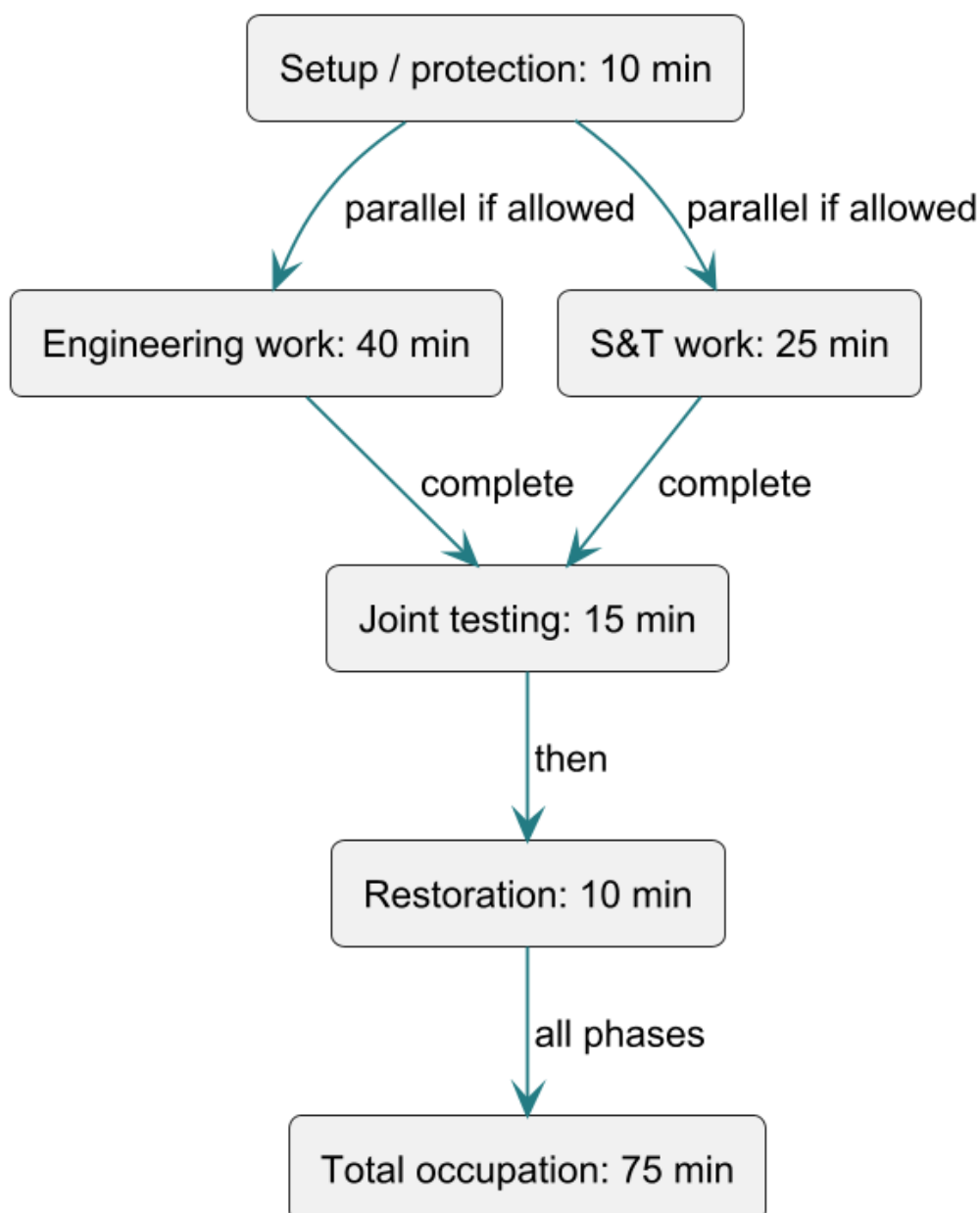
Different departments describe the same railway differently.

Map chainage, routes, points and electrical sections to versioned operational footprints. Ambiguous geography must block unsafe grouping rather than be guessed.

D20 / Example package execution DAG

Illustrative TEST recipe only: $10 + \max(40,25) + 15 + 10 = 75$ minutes, permitted only with validated concurrency and separate qualified resources. With one shared exclusive crew and sequential work, the same phases require 100 minutes. These values are not Railway rules.

D20 - Example package execution DAG



Count the complete occupation, including restoration.

Illustrative arithmetic: $10 + 40 + 25 + 15 + 10 = 100$ minutes sequentially; $10 + \max(40,25) + 15 + 10 = 75$ when parallel work is allowed. The 25 minutes is an example, not a measured field saving.

What the winner research changes

A targeted review, with clear evidence limits.

Three archived decks inspected

Cannon Crew, Framework Fanatics and SMARTIrrIS show the value of section coverage, visible workflows and risk/mitigation mapping. Their individual winner status was not independently verified. Dense labels and unsupported benefit figures were not adopted.

Two verified winner accounts

MathWorks accounts for Solar Masters (2024) and TwinX (2025) emphasize a real workflow gap, working modules, iteration, testing and practical use. Their original submission decks were not obtained. These accounts do not establish that a visual style causes selection.

Our resulting approach

Lead with a railway planning decision. Show the named source systems, complete package logic, independent checker and human approval. Use one transparent arithmetic example instead of an unmeasured savings claim.

An honest innovation claim

Joint maintenance planning already exists in railway guidance. The proposed contribution is an integrated, explainable and auditable implementation with compatibility evidence, whole-package checks and stable recovery. Novelty must be assessed against real operational workflows.

Pilot measurement

Compare the same task set and constraints against an agreed baseline. Measure required-work coverage, total occupation, unexplained omissions, changed assignments, solve time and checker violations. Report failed or timed-out cases, not only successful runs.

Sources and further detail

Primary sources, reference decks and the project research trail.

Official SIH2026 problem statement

<https://www.sih.gov.in/sih2026PS>

Railway joint rolling block guidance

https://www.iricen.gov.in/iricen/other_manual/ircm_ref_docs/1801.pdf

CRIS: TMS

<https://cris.org.in/loadpage?page=proTMS>

CRIS: TDMS

<https://cris.org.in/loadpage?page=proTDMS>

CRIS: COA

<https://cris.org.in/loadpage?page=proCOA>

Google OR-Tools: CP-SAT

https://developers.google.com/optimization/cp/cp_solver

Public archived presentation collection

<https://github.com/DgrnBoi/SIH-Past-PPTS-2024>

MathWorks: verified competition winners

<https://www.mathworks.com/academia/students/competitions/hackathons/winners.html>

Full project design

NIRVIKALP_SAMARATH_Production_Blueprint.pdf: 98-page source blueprint. The original PlantUML models supply this companion's 22 diagram pages. Diagrams are design specifications, not screenshots proving implementation.

Detailed research and speaking preparation

Research_Analysis_and_Judge_Guide.md includes the winner-account links, deck-by-deck analysis, source limits, requirement mapping, speaking notes and ten judge questions.

Before submission

Fill the registered Team ID and public demo link in the editable PowerPoint, then export an updated six-page PDF. Check link access without login. Do not upload this technical companion as extra submission slides.